

Autonomous by Design — AI Agents for Multi-Step Task Automation

Theme: Build intelligent systems capable of automating workflows, making decisions, interacting with tools, and executing multi-step tasks with minimal human intervention — exploring the convergence of AI agents, automation frameworks, APIs, and intelligent software systems.

DOMAIN	CATEGORY	PS TYPE
AI Agents & Automation	SOFTWARE	Open Innovation

BACKGROUND

Most software today still waits to be told exactly what to do — one function call, one API request, one click at a time. But real work rarely comes as a single step: it involves gathering information, deciding what to do next, calling the right tool at the right moment, recovering from failures, and following through on a multi-step goal. AI agents — systems that can reason, plan, and act using tools and APIs — offer a way to automate this kind of work end-to-end, rather than automating only isolated pieces of it.

PROBLEM STATEMENT

Design and build an AI agent-based system that can take a high-level goal, break it into a sequence of steps, decide which tools or APIs to invoke at each step, and carry the task through to completion with minimal human input — while remaining transparent, controllable, and safe to use in a real workflow.

EXPECTED FOCUS AREAS

- Planning and decomposing a high-level goal into executable multi-step actions
- Memory and context management across long-running or multi-turn tasks
- Human-in-the-loop checkpoints for oversight on sensitive or irreversible actions
- Tool-use and API orchestration, including chaining multiple tools together
- Decision-making under uncertainty, with graceful error handling and retries
- Observability into what the agent decided, why, and what it executed

EXPECTED OUTCOME

A working agentic system that, given a realistic high-level task (such as researching a topic, managing a workflow, or coordinating across multiple tools/APIs), autonomously plans and executes the necessary steps to completion with little to no manual intervention.

Solutions will be evaluated on reasoning quality, reliability across multi-step execution, sensible use of tools and APIs, transparency of the agent's decisions, and safe handling of failures or ambiguous situations.

IMPACT

PRODUCTIVITY	ADAPTABILITY	SCALABLE AUTONOMY
Frees people from repetitive multi-step work by letting agents handle it end-to-end.	Agents can respond to changing conditions and tool outputs, unlike fixed automation scripts.	Lays the groundwork for intelligent systems that reliably act on our behalf at scale.